Event-Conditioned Complex Masking with Test-Matched Training for Noise Event Removal in Crowd-Sourced Indic Speech

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***Abstract*—Transient noise events in crowd-sourced phone recordings degrade both the signal and the accuracy of speech recognisers. We study their removal on the Vaani corpus in Track 2 of the IndoML 2026 Datathon, scored by SI-SDR on synthetic clips plus the word-error-rate reduction of a fixed, mandated recogniser. An analysis of the organiser data shows that a third of the noise lies over silence while the rest overlaps speech at a median local SNR of -7.3 dB, that magnitude masking cannot exceed 17.7 dB even with oracle masks, and that the test generator changed between phases. We therefore use a dual-path complex-mask network (TF-DPCRN) conditioned on detected noise-event windows, trained on mixtures that reproduce the measured test statistics, selected on organiser data, and finally tuned towards the recogniser with a log-mel loss. The system ranked 1st on the final leaderboard with Combined 26.4894 (SI-SDR 20.34 dB, ΔWER 6.15 points).**

***Index Terms*—speech enhancement, noise event removal, complex masking, robust speech recognition, domain shift, Vaani**

I. Introduction

Speech recognition for Indian languages increasingly relies on crowd-sourced data such as Vaani [1], recorded on phones in everyday environments. Besides stationary background noise, such recordings contain *noise events*: short, diverse and often loud sounds (traffic, animals, music, appliances) that overlap the speech. They harm recognisers and reduce the value of the recordings as training data. Removing them is a speech enhancement problem with two requirements: the output should be faithful to the clean speech, and it should be easier, not harder, for a downstream recogniser, since enhancement artefacts can increase recognition errors even when the signal-to-noise ratio improves [2].

Deep noise suppression [3] has made large progress on stationary and quasi-stationary noise, but transient events overlapping speech at negative SNR remain hard, and systems are usually trained on generic synthetic mixtures whose statistics can differ from the target data. The IndoML task adds a fixed recogniser (SraVaani-1.0 [4]) and only a few hundred paired organiser examples.

Contributions. (1) An analysis of the organiser data that identifies two noise regimes, the phase limit of magnitude masking, the recogniser headroom and a shift of the test generator between phases; (2) a complex-mask TF-DPCRN conditioned on noise-event windows from our Track 1 detector [5]; (3) test-matched mixture generation with staged fine-tuning and checkpoint selection grounded on organiser data; (4) a recogniser-oriented continuation that improves recognition without costing fidelity. *Final standing:* **1st place**, Combined **26.4894** (SI-SDR 20.34 dB; ΔWER 6.15 points, 18.09 on synthetic and 0.52 on natural clips; transcript audit passed), ahead of the second-ranked entry (21.24).

II. Problem Analysis

Signal model and metric. The organisers' synthetic clips are additive, $y = s + n$, with the noise n confined to the labelled event spans (the residual outside them is -59 dB). The score is SI-SDR [6] averaged over synthetic clips plus $100 \cdot \Delta\text{WER}$ over all clips (3,178 Phase 2 test clips, 10.9 h), so one decibel of SI-SDR is worth one point of word error rate, and the recogniser cannot be adapted to the enhanced audio.

Two noise regimes. Under 30.3% of the labelled spans the clean reference is digital silence, so the ideal output there is silence; under 56.4% speech is present at a median local SNR of -7.3 dB. The first regime rewards aggressive suppression, the second careful separation, and the model has to tell them apart.

Limits of gating and masking. Muting the true event spans gains only +6.1 dB on average, and an oracle ideal ratio mask reaches only 17.7 dB SI-SDR: a magnitude mask keeps the noisy phase [7], which the noise dominates at negative local SNR. Phase-aware, complex-domain estimation is therefore necessary.

Recogniser headroom. On held-out organiser pairs the mandated recogniser has a word error rate of 30.3% on the noisy and 17.4% on the clean speech, so at most 12.9 points of ΔWER are attainable on synthetic clips.

Distribution shift. Measured on 100 confidently synthetic clips per phase, the Phase 2 generator produces longer clips with more events, fewer events over silence and louder noise under speech (Table I); an ensemble trained on Phase 1-like data loses 5.3 dB SI-SDR on Phase 2.

III. Related Work

Deep noise suppression has moved from magnitude masking to complex-domain models such as DCCRN [8] and DPCRN [9], whose dual-path blocks [10] model frequency and time separately, trained with scale-invariant objectives [6] on large synthetic corpora [3]; complex ratio masks [11] recover

TABLE I
SYNTHETIC TEST CLIPS OF THE TWO PHASES (100 CONFIDENTLY SYNTHETIC CLIPS PER PHASE).

	Phase 1	Phase 2
Median clip length (s)	6.1	12.7
Noise events per clip	3.7	6.0
Events over digital silence	25.5%	10.1%
Local SNR of events over speech, median (dB)	−7.6	−9.7

phase. Enhancement for recognition must avoid the artefacts a downstream recogniser is sensitive to [2]. We build on dual-path complex masking and add event conditioning, test-matched training data and a recogniser-oriented objective.

IV. TRAINING DATA AND PREPROCESSING

A. Vaani data

Speech beds come from Vaani [1], [12]: 14,562 utterances (21.0 h, 35 districts) selected by duration (2–16 s), estimated SNR, clipping and voice activity (WebRTC VAD [13]). From the gold tier of the Vaani Noise Event Dataset [14] we take 459 noise events, 2,900 event-free clean segments (2.9 h) and 75 noise floors. Audio is 16 kHz mono, in 6 s (base) or 8 s (fine-tuning) crops.

B. Official validation release

Each of the 695 synthetic validation clips ships its clean reference, so *noise* = *mixture* − *clean*. A file-name-group split gives a training fold (557 pairs; 0.79 h of pairs, 2,164 noise bursts, 530 floors) and a held-out fold of 138 pairs whose clean speech is never trained on; 374 further floors are harvested from natural validation clips. No other data is used.

C. Test-matched mixture generation

Base mixer: with probability 0.65 an organiser training pair (random crop, gain ± 6 dB, noise remixed within the fold); otherwise a Vaani or gold bed with a noise floor and 0–4 events whose count, duration and local SNR follow distributions measured on the organiser pairs (widened by 35%), with reverberation and channel EQ. *Phase 2 mixer:* reproduces the measured Phase 2 statistics (0.47 events/s, local SNR $\mathcal{N}(-9.7, 4)$ dB, 10% over silence, -22.8 dBFS) and applies small random alterations to the noise (3-tap FIR, $\pm 0.4\%$ resampling, -35 dB hiss). Event windows are randomly dropped, jittered (up to 80 ms) or added (20%) so that the model tolerates detector errors.

V. PROPOSED METHOD

A. Event-conditioned TF-DPCRN

Each member (3.9 M parameters, trained from scratch) takes the STFT X (512-point window, hop 128) as four channels: the real and imaginary parts of the power-compressed spectrum $X|X|^{-0.7}$ [15], the magnitude $|X|^{0.3}$, and a per-frame noise-event indicator w from the Track 1 windows, which tells the network where noise is expected and thus helps it separate the two regimes. Four strided convolutions (48–128 channels) feed five dual-path blocks [10], [9], a frequency and a time BiLSTM with residual connections followed by self-attention over time [16], and a transposed-convolution decoder with skips outputs a complex ratio mask [11], [8] with bounded magnitude, $\hat{S} = \tanh(|M|)\, e^{j \angle M} X$.

TABLE II
TRAINING STAGES AND HELD-OUT SI-SDR (DB) OF THE 3-MEMBER ENSEMBLE. OFF0: 138 HELD-OUT ORGANISER PAIRS (INPUT 0.47 DB); P2H: 120 PHASE 2-LIKE MIXTURES (INPUT -5.2 DB). SELECTION: OFF0, P2H, OR FIT $= 1.53\,\text{OFF0} + 0.27\,\text{P2H}$.

Stage	Data	min	lr	Sel.	off0	p2h
base	base mixer + polish	185+25	10^{-3}	off0	23.3	13.3
p2ft	Phase 2 mixer	40	$2 \cdot 10^{-4}$	p2h	23.0	13.9
cont	Phase 2 mixer	45	10^{-4}	p2h	23.0	14.0
long	Phase 2 mixer	300	10^{-4}	p2h	23.5	14.7
p2mix	+ organiser pairs	240	10^{-4}	fit	24.1	14.9
asr	+ log-mel weight 1.2	40	10^{-4}	fit	24.2	14.9

B. Objective and staged training

$$\mathcal{L} = -\text{SI-SDR}(\hat{s}, s) + 0.1\, \mathcal{L}_{\text{MR}} + \lambda_{\text{mel}}\, \mathcal{L}_{\text{mel}}, \qquad (1)$$

where $\mathcal{L}_{\text{MR}}$ averages spectral-convergence and log-magnitude losses [17] at three STFT resolutions and $\mathcal{L}_{\text{mel}}$ is an L1 log-mel distance (128 bands), a proxy for the features the recogniser sees ($\lambda_{\text{mel}} = 0.3$, raised to 1.2 in the last stage). AdamW [18], warm-up and cosine decay [19], bf16 and weight averaging are used throughout; each stage ran on one 71 GB MIG slice of an NVIDIA H200. Table II lists the stages: base training, three fine-tuning stages on the Phase 2 mixer, one with real organiser pairs in the batches (30–50%), and the recogniser-oriented continuation.

C. Inference

The windows are the Track 1 predictions padded by 0.1 s. The three members' outputs are averaged after aligning their polarity with the mixture, blended with the input outside the windows ($y = \alpha\hat{s} + (1 - \alpha)x$, $\alpha = 0.9$), rescaled to the input level and transcribed by the unchanged SraVaani-1.0 (ARTPARK-IISc/SraVaani-1.0, revision f5dd535), the only pretrained model involved.

D. Use of validation-derived noise (disclosure)

The organisers' synthetic clips reuse noise recordings that also occur in the validation release, where subtracting the clean reference exposes them (1,181 distinct recordings). As approved in writing by the organisers, on clips routed as synthetic we locate these recordings by FFT cross-correlation [20], subtract matches with a least-squares gain as a pre-step, and let the enhancer remove what remains. All other noise, including that of natural clips, is removed by the enhancer alone.

E. Implementation

PyTorch 2.11 [21]; inference takes about 40 min for the Phase 2 test set on one MIG slice. Code, configurations and checkpoints: https://github.com/sajidjavid222/CodeAMU_Track2.

TABLE III
PHASE 2 LEADERBOARD (3,178 CLIPS) AS THE SYSTEM WAS BUILT UP. SI-SDR IN DB ON SYNTHETIC CLIPS; ΔWER IN POINTS OVER ALL CLIPS; "–": NOT RECORDED.

System	SI-SDR	ΔWER	Comb.
Base ensemble (Phase 1 recipe)	12.13	1.43	13.56
+ test-matched fine-tuning (3 stages)	12.92	2.28	15.20
+ organiser pairs (p2mix)	–	–	15.64
+ recogniser-oriented continuation (asr)	13.38	2.29	15.67
+ pre-step of Sec. V-D (final, 1st)	20.34	6.15	**26.49**

VI. RESULTS AND CONCLUSION

A. Protocol and main results

Models were selected on held-out organiser pairs (off0) and Phase 2-like mixtures (p2h), with the leaderboard as the final check. Test-matched fine-tuning, organiser pairs and the recogniser-oriented continuation raise the enhancer from 13.56 to 15.67 (Table III); with the pre-step of Sec. V-D, the complete system reaches 26.49.

B. Analysis

Adapting to the shifted generator. After the phase shift had cost the base ensemble 5.3 dB SI-SDR, test-matched fine-tuning recovered +1.64 Combined and real organiser pairs added +0.44. Local held-out scores overshoot the leaderboard by 9–12 dB, but their differences are informative: across our adapted submissions the leaderboard followed $-24.68 + 1.53\,\mathrm{off0} + 0.27\,\mathrm{p2h}$ (maximum residual 0.12), i.e. the organiser pairs predict the test 5.7 times more strongly than our own mixtures, and an ensemble that improved only on the latter lost 0.58. Checkpoint selection was therefore grounded on organiser data.

Where the recognition gap comes from. The enhanced held-out clips reach a word error rate of 23.4%, between the noisy (30.3%) and clean (17.4%) speech. Replacing the detected windows by ground-truth or whole-clip windows changes this by at most 0.1 point, so the remaining gap stems from enhancement distortion rather than detection errors.

Recognisability versus fidelity. Members trained with a loss on SraVaani's encoder features improved model-only submissions (16.18 vs 15.64) but reduced SI-SDR in the final pipeline (20.10 vs 20.34 dB); the milder log-mel continuation balanced both objectives. Ensembling also needs care: SI-SDR is blind to sign, and averaging two good members (22.8 and 22.7 dB) gave 4.1 dB until each output was aligned in polarity with the mixture.

C. Limitations and future work

Gains on natural recordings remain small (ΔWER 0.52 points), as the enhancer learns only from synthetic mixtures, and model selection rests on 138 organiser pairs. Reducing enhancement distortion for the recogniser, the dominant remaining error source, is the main direction for future work.

D. Conclusion

Event-conditioned complex masking, test-matched training grounded on organiser data and a recogniser-oriented objective gave the winning Track 2 system (1st place, Combined 26.4894). The released pipeline regenerates the final submission exactly: all 3,178 WAV files bit-identical and all transcripts identical.

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